# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE CODE:** MLA03 **COURSE NAME: Reinforcement learning for Environment and Agent**

**COURSE OUTCOME COVERED**

***CO3:*** *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

*Assessment Tool Weightage: 25%*

Dr G. A. SENTHIL

**QUESTIONS: 10** **Total Marks: 50**

Annexure A

**Data Interpretation** **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent. **(5 Marks)**

A.The reward obtained by the Reinforcement Learning (RL) agent increases from 10 to 45 over multiple training episodes. This indicates that the agent is learning from its interactions with the environment and gradually improving its policy.

Calculation

Initial Reward = 10
Final Reward = 45

1. Absolute increase in reward

Increase=Final Reward−Initial Reward =45−10=35

So, the reward increased by 35 points.

2. Percentage improvement

Percentage Improvement=Initial RewardFinal Reward−Initial Reward×100 =1045−10×100 =1035×100 =350%

Thus, the final reward is 350% higher than the initial reward.

Alternatively, the final reward is:

1045=4.5

So, the agent's final reward is 4.5 times the initial reward.

Interpretation

1. Positive learning trend: The gradual increase from 10 to 45 shows that the agent is learning from its previous actions and experiences.

2. Policy improvement: Higher rewards indicate that the agent's policy is becoming more effective at selecting actions that produce better outcomes.

3. Better decision-making: As training progresses, the agent learns to avoid poor actions and select actions that provide higher rewards.

4. Learning performance: The increasing reward trend indicates improved learning performance and successful adaptation to the environment.

5. Convergence indication: If the reward eventually becomes stable around 45, it may indicate that the policy is approaching a good or stable solution.

Conclusion

The reward increase from 10 to 45, representing a 35-point increase or 350% improvement, demonstrates that the RL agent is progressively learning and improving its policy. Therefore, the increasing reward trend is a positive indicator of successful training and improved decision-making performance.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

A. Data Interpretation – Epsilon-Greedy Agent

Given Data

| Epsilon (ε) | Exploration | Exploitation | Cumulative Reward |
|---|---|---|---|
| 0.1 | 10% | 90% | Highest |
| 0.3 | 30% | 70% | Medium |
| 0.5 | 50% | 50% | Lowest |

Explanation

The epsilon-greedy algorithm balances exploration and exploitation.

- Exploration: The agent tries different actions to discover potentially better actions.
- Exploitation: The agent selects the action that it currently believes gives the highest reward.

The value of ε (epsilon) determines the probability of exploration.

Interpretation

$\varepsilon = 0.1$

Exploration=10% Exploitation=90%

The agent explores only 10% of the time and exploits the best-known action 90% of the time. Therefore, it makes more decisions based on its learned knowledge and obtains the highest cumulative reward.

$\varepsilon = 0.3$

Exploration=30% Exploitation=70%

The agent explores more than when $\varepsilon = 0.1$. This can help discover new actions, but it also means that the agent sometimes chooses actions that are not currently optimal. Hence, the cumulative reward is lower than $\varepsilon = 0.1$.

$\varepsilon = 0.5$

Exploration=50% Exploitation=50%

The agent spends half of its decisions exploring random actions. As a result, it exploits the best-known actions less frequently, leading to the lowest cumulative reward among the three values.

Conclusion

The relationship can be summarized as:

$\epsilon\uparrow \Rightarrow \text{Exploration}\uparrow$ $\epsilon\uparrow \Rightarrow \text{Exploitation}\downarrow$

Since the highest cumulative reward occurs at $\varepsilon = 0.1$, it provides the best performance among the tested values.

$\epsilon=0.1$ is the best-performing value

Reason: It provides a good balance in this experiment, with 90% exploitation and 10% exploration, allowing the agent to make more rewarding decisions while still performing some exploration.

**(5 Marks)**

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.
**(5 Marks)**

A.Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared at 100, 200, and 300 episodes.
The observation is:
Q-Learning consistently gives slightly higher average rewards than SARSA.
Interpretation

| Episodes | Q-Learning | SARSA | Observation |
|---|---|---|---|
| 100 | Higher | Slightly lower | Q-Learning performs better |
| 200 | Higher | Slightly lower | Q-Learning maintains the lead |
| 300 | Higher | Slightly lower | Q-Learning still performs better |

Since exact reward values are not provided, the numerical performance difference cannot be calculated. We can only interpret the observed difference qualitatively.

Performance Difference

The performance difference at any episode is:

Performance Difference=Q-Learning Reward−SARSA Reward

Because Q-Learning has a slightly higher reward:

Q-Learning Reward>SARSA Reward

Therefore, Q-Learning has a slight performance advantage over SARSA throughout the experiment.

Convergence

Convergence means that the algorithm's learned policy/reward becomes relatively stable as training progresses.

Since Q-Learning consistently obtains higher rewards at 100, 200, and 300 episodes, the observed results suggest that Q-Learning converges faster or reaches a better-performing policy earlier than SARSA.

However, strictly speaking, reward values alone at only three episode points are not sufficient to prove convergence speed. A reward-vs-episode graph showing when each algorithm stabilizes would provide stronger evidence.

Why Q-Learning May Perform Better

- Q-Learning is an off-policy algorithm and learns using the maximum estimated future reward.
- SARSA is an on-policy algorithm and learns according to the action actually selected by its current policy.
- Therefore, Q-Learning can sometimes learn a more rewarding policy faster, particularly in environments where the optimal action has a clear advantage.

Final Answer

Q-Learning consistently achieves slightly higher average rewards than SARSA at 100, 200, and 300 episodes. Therefore, Q-Learning shows better performance in the observed experiment. Based on the consistently higher rewards, Q-Learning appears to converge faster and reach a better-performing policy earlier than SARSA. However, exact convergence speed requires the reward-versus-episode curves or stabilization points for both algorithms.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

**A.**Data Interpretation – Discount Factor in Reinforcement Learning

Given

The RL agent is tested with different discount factor values:

[
\gamma = 0.2,;0.5,;0.9
]

The average reward increases as the discount factor increases.

Meaning of Discount Factor

The discount factor, represented by $\gamma$ (gamma), determines how much importance the agent gives to future rewards compared with immediate rewards.

The value of $\gamma$ lies between 0 and 1.

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Interpretation

| Discount Factor ($\gamma$) | Importance of Future Rewards | Expected Learning Behavior |
|---|---|---|
| 0.2 | Low | Focuses mainly on immediate rewards |
| 0.5 | Moderate | Considers both immediate and future rewards |
| 0.9 | High | Strongly considers long-term rewards |

1. $\gamma = 0.2$

The agent gives low importance to future rewards.

It mainly focuses on the reward received immediately after an action. This can result in short-sighted decisions and, according to the given observation, produces the lowest average reward.

2. $\gamma = 0.5$

The agent considers both immediate and future rewards.

It provides a moderate balance between short-term and long-term decision-making. Therefore, the average reward is higher than with $\gamma = 0.2$.

3. $\gamma = 0.9$

The agent gives high importance to future rewards.

It learns to select actions that may provide smaller immediate rewards but lead to greater rewards in the future. Since the question states that average reward increases with $\gamma$, $\gamma = 0.9$ gives the highest average reward.

Effect on Long-Term Reward

The discounted return can be represented as:

[
G_t=R_{t+1}+\gamma R_{t+2}+\gamma^2R_{t+3}+\cdots
]

For example, consider a reward received 3 steps later:

- For $\gamma = 0.2$:

[
(0.2)^3=0.008
]

- For $\gamma = 0.5$:

[
(0.5)^3=0.125
]

- For $\gamma = 0.9$:

[
(0.9)^3=0.729
]

Thus, $\gamma = 0.9$ gives much greater importance to future rewards than $\gamma = 0.2$.

Conclusion

Since the average reward increases as γ increases, the most suitable value among the tested values is:

[
\boxed{\gamma=0.9}
]

Reason: γ = 0.9 gives greater importance to long-term rewards, allowing the agent to make decisions that maximize future returns. Therefore, based on the observed results, γ = 0.9 provides the best learning performance.

5-Mark Answer: A higher discount factor makes the agent more future-oriented. γ = 0.2 emphasizes immediate rewards, γ = 0.5 provides a balance, and γ = 0.9 strongly considers future rewards. Since the average reward increases with γ, γ = 0.9 is the most suitable value and gives the best performance in this experiment.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

A.Given Data

- Initial success rate = 45% at 50 episodes
- Final success rate = 92% at 200 episodes

1. Absolute Improvement

Improvement=Final Success Rate−Initial Success Rate =92%−45% =47 percentage points

So, the success rate improved by 47 percentage points.

2. Percentage Improvement

Percentage Improvement=4592−45×100 =4547×100 =104.44%

Therefore, the success rate increased by approximately 104.44% relative to its initial value.

3. Interpretation of the Trend

| Episodes | Success Rate |
|---|---|
| 50 | 45% |
| 200 | 92% |

The success rate shows a strong positive trend as the number of training episodes increases.

- At 50 episodes, the agent succeeds in only 45% of cases, indicating that the policy is still developing.
- At 200 episodes, the success rate reaches 92%, showing that the agent has learned much better action-selection strategies.
- The large increase indicates that the agent is effectively learning from its interaction with the environment.

Has the Agent Learned an Optimal Policy?

A 92% success rate is strong evidence that the agent has learned an effective policy, but it does not prove that the policy is mathematically optimal.
To claim an optimal policy, we would normally need additional evidence such as:

- The success rate becoming stable over further episodes.
- Comparison with the theoretically optimal policy or benchmark.
- Stable/high rewards over a longer training period.
- Performance on unseen test episodes.

Conclusion
The RL agent's success rate increased from 45% to 92%, an improvement of 47 percentage points (104.44% relative improvement). This demonstrates substantial learning and policy improvement. The 92% success rate suggests that the agent has learned a highly effective policy, but based only on the given data, we cannot conclusively say that it is optimal. More training data and convergence evidence would be required.
Effective policy: Yes | Proven optimal policy: No

**(5 Marks)**

***Code of Conduct:***

*I, K.Mounith(192425090)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** K.Mounith

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Data | 25% | Accurately interprets all data patterns, trends, and relationships | Interprets data correctly with minor gaps | Partial understanding; misses key trends | Misinterprets or fails to understand data |
| Analysis | 25% | Provides deep analysis with strong logical reasoning and justification | Provides reasonable analysis with some justification | Basic analysis with weak reasoning | No meaningful analysis provided |
| Application of Concepts | 20% | Effectively applies relevant concepts/tools to interpret data accurately | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply concepts to data |
| Conclusion | 20% | Draws clear, meaningful conclusions with strong insights and implications | Provides valid conclusions with moderate insights | Conclusions are vague or weak | No clear or valid conclusions |

| | | | | | |
|---|---|---|---|---|---|
| Clarity | 10% | Presents interpretation clearly using proper structure, visuals, and terminology | Generally clear with minor issues | Lacks clarity or proper structure | Poor presentation and difficult to understand |